



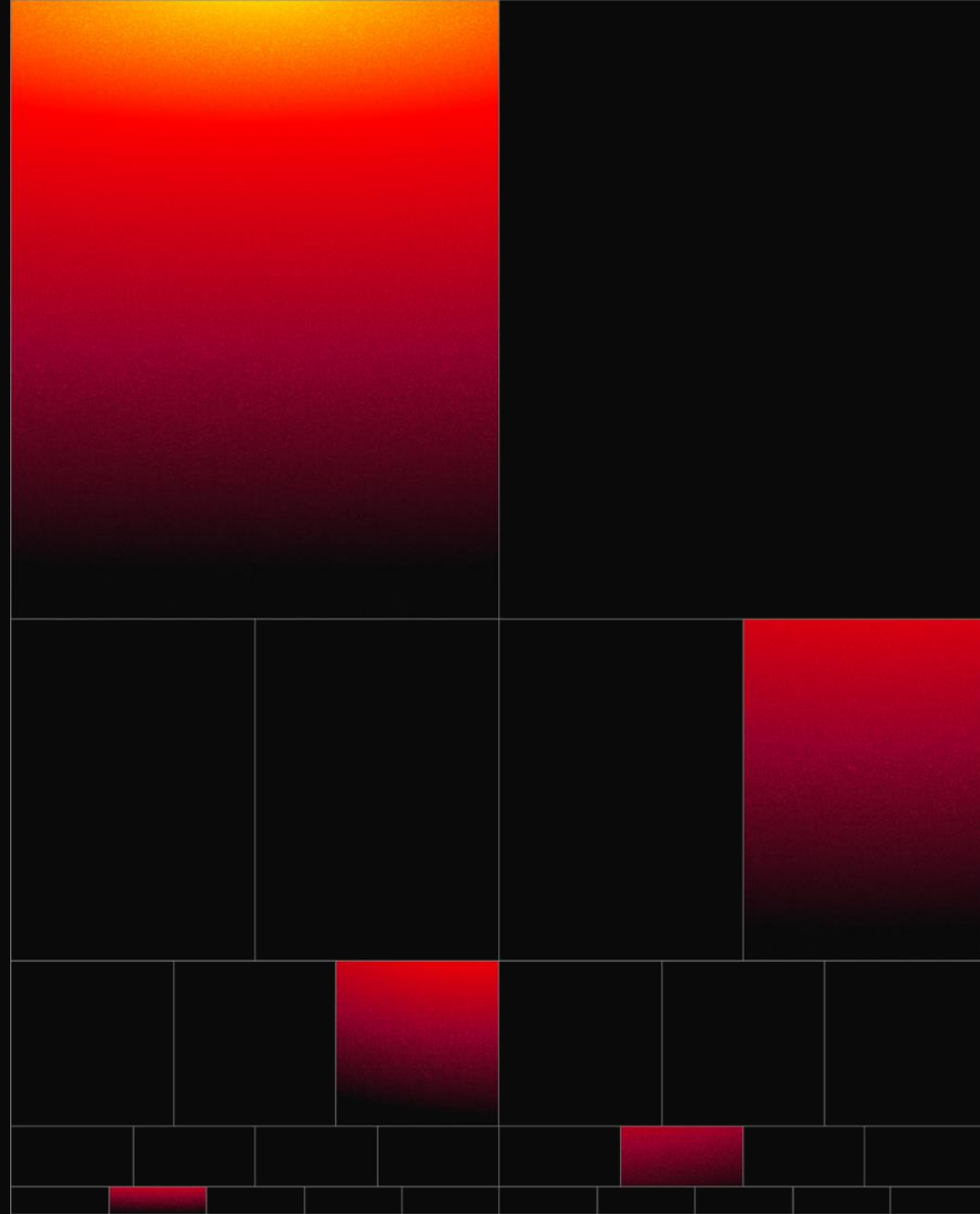
Gurobi Intelligence

GenAI in Gurobi

[WWW.GUROBI.COM](https://www.gurobi.com)

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Behind Every Great Decision



Agenda

01 Intelligence Hub

02 Gurobot

03 Modeler

04 Explainer

We are live: intelligence.gurobi.com

Welcome to the Gurobi Intelligence Hub

Select an agent to get started:

 **Gurobot**

Instant, expert-level Gurobi support—resolving issues faster with personalized, context-aware answers, code snippets, and documentation links. Gurobot also helps you create a support ticket if needed.

[Read more ↗](#)[Launch Gurobot >](#)

 **Modeler** Beta

Guides business users from problem description to validated, production-ready gurobipy code—through targeted questions, incremental specification building, and automated testing backed by Gurobi’s knowledge base and best practices.

[Read more ↗](#)[Launch Modeler >](#)

 **Explainer** Experimental

Helps users diagnose and understand optimization models faster by combining automated model analysis, Gurobi’s curated knowledge base, and relevant business context.

[Read more ↗](#)[Launch Explainer >](#)

 **Recent Chats** Last 10 chats

 Make my problem feasible by reducing the amount of EXPLAINER 6d ago 

 moodel.ilp EXPLAINER 27d ago 

 infeasible_exp_sos_iis.ilp EXPLAINER 08/05/2026 

 **Access Agents from Your IDE or Coding Assistant** Experimental

MCP - LOCAL SERVER

Install the local MCP server to use Gurobot and Explainer inside your favorite IDE extension or AI coding assistant — powered by the Intelligence Hub backend.

[Read more ↗](#)[VIEW ALL >](#)

Intelligence Hub

All our Gen AI tools in one place

The Intelligence Hub is where users access Gurobi's AI agents — either directly in the web UI or through API-key authenticated MCP access.



Home to Gurobi's AI agents

Gurobot, Explainer, and Modeler in one shared experience



Grounded in Gurobi Expertise

Answers draw from curated documentation, examples, best practices, and expert-authored content.



Secure workspace

Each chat has isolated file handling, generated artifacts, and deletion controls



One usage model

Daily token limits are shared across all agents

Security & Privacy

How the Intelligence Hub handles your data

✓ Never used to train LLMs

Your prompts, files, outputs, and chat history are never used to train or fine-tune any large language model.

✓ Isolated by design

Each session runs in a dedicated environment — no shared state or filesystem with any other user or organization.

✓ Encrypted everywhere

HTTPS in transit; application-layer encryption at rest with a unique key per file.

✓ You control deletion

Delete one chat, an agent's history, or everything — from Settings, at any time, permanently.

✓ Sandboxed MCP server

Limited to the directory you mount; credentials live only in environment variables; activity visible in standard Docker logs.

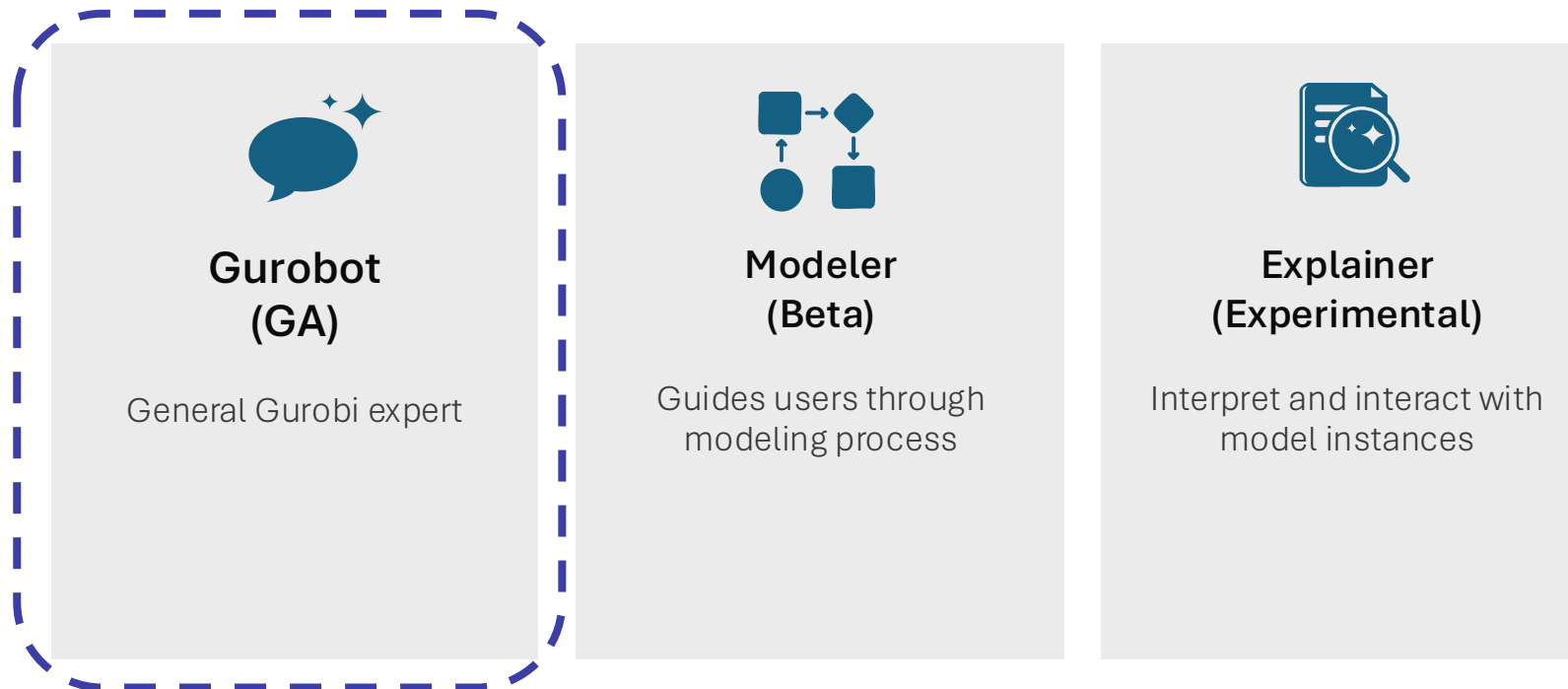
✓ Keep personal data out

Don't include data that identifies a person — names, addresses, IDs, payment details — in prompts or files.

Generative AI can make mistakes — verify output before relying on it. Binding terms: the Intelligence Hub [Terms of Service](#).

Agents

Overview of our agents



Beta Features:

- Further along in development
- Still refining how they work based on what we learn

Experimental Features:

- Earlier in development
- Give users a first look at something we're exploring
- More likely to change significantly

Gurobi values exceptional support

- Gurobi Expert Team:
 - 30 PhD Level Experts
 - 10 countries
- Tickets with Commercial customers
- Attend to our Community Portal
- Curate over 500 knowledge documents:
 - Documentation
 - Articles
 - Modeling Examples
 - Coding Examples
- Gurobot provides instant frontline support





Your always-on Gurobi Expert

An AI support assistant for Gurobi product questions, licensing context, troubleshooting, modeling tricks, code generation, and documentation-backed answers



Context aware support

Knows who you are and what licenses you have



Documentation-backed answers

Responses include links to relevant Gurobi knowledge



Code help and review

Generate, rewrite, debug, and validate Gurobi code



Escalation to experts

Turn a chat into a support request with an AI-generated summary and classification

Hi, I'm Gurobot. How would you like to get support?



Chat with Gurobot

Chat with our AI Gurobot to get instant assistance

Start chatting below



Gurobi Expert Team

Open a support request to contact the Gurobi Expert Team.

Contact Expert Team

Type your message...





The **MIPGap** is a relative optimality tolerance parameter used in Mixed Integer Programming (MIP) solvers. It is defined as:

$$\text{gap} = \frac{|z_P - z_D|}{|z_P|}$$

where:

- z_P is the **primal bound** (the incumbent/best feasible solution found)
- z_D is the **dual bound** (the best proven lower/upper bound from the branch-and-bound tree)

Gurobi terminates with an optimal result when this gap falls below the MIPGap threshold. The default value is 10^{-4} (i.e., 0.01%).

Would you like to know more about how to tune this parameter or about the related MIPGapAbs parameter?

Learn More:

- [What is the MIPGap?](#) ★★☆☆
- [What's New - Gurobi 13.0 | Gurobi](#) ★★☆☆
- [Parameter Reference](#) ★★☆☆
- [Gurobi 13.0 released](#) ★★☆☆
- [From Speed to Scalability: Why Gurobi 13.0 Is Built for Today's Optimization Demands | Gurobi](#) ★★☆☆
- [Why are there large or increasing MIP gap values?](#) ★☆☆☆



Create a support request

Type your message...



Smart Links

Easy Escalation

Feedback Mechanism

Please use!

We use these to make improvements

Autogenerated Escalations

Create a Support Request

Support Category

Modeling

Title

MIPGap Definition and How to Set It in gurobipy

Description

This support request is created on behalf of the user:

The user asked about the definition of MIPGap, specifically what it means in the context of Mixed Integer Programming.

The user then asked how to set the MIPGap parameter in Gurobi Python (gurobipy) code.

Initial question: Answer the following question very concisely

☐ Our production system is down
Check only for EMERGENCY production system interruptions affecting end users (not developers)

BACK TO CHAT SUBMIT

Generate and run gurobipy code

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what happens if you run Compute IIS on an unbounded problem

Based on the knowledge base information, here's what happens when you try to compute an IIS on an unbounded problem:

IIS Computation on Unbounded Models

Compute IIS only works on infeasible models. If you attempt to run `computeIIS()` on an unbounded problem, you will encounter issues because IIS (Irreducible Inconsistent Subset) computation is specifically designed to identify the subset of constraints causing **infeasibility**, not unboundedness.

What Actually Happens

- INF_OR_UNBD Status:** Sometimes after optimization, Gurobi returns an `INF_OR_UNBD` (infeasible or unbounded) status. In this case, you **cannot reliably compute an IIS** because the model might be unbounded rather than infeasible. Some presolve steps prevent Gurobi from distinguishing between the two conditions.
- Error Message:** If you try to compute an IIS when the model is actually unbounded (or feasible), you'll get an error message: **"Cannot compute IIS on a feasible model"** or similar indication that IIS computation is not applicable.

How to Proceed

To determine whether your model is truly infeasible or unbounded, follow these steps:

- Set the `DualReductions` parameter to 0:**

```
python
model.setParam('DualReductions', 0)
```

This conversation is from a previous version of our platform and has been closed. If you'd like to continue, please start a new chat.

The Gurobi AI Agent is using generative AI and may produce inaccuracies. By using this service, you agree to our [Privacy Policy](#) and [Terms of Service](#).
You have used 3% of your daily limit.

Documentation Downloads Support Steven Edwards

Chat files

unbounded_iis_demo.py (4.3 KB)

```
import gurobipy as gp
from gurobipy import GRB

# Create an unbounded model
print("=" * 70)
print("DEMONSTRATION: Computing IIS on an Unbounded Problem")
print("=" * 70)

with gp.Env(empty=True) as env:
    env.setParam('OutputFlag', 1)
    env.start()

# Create a simple unbounded model
model = gp.Model("unbounded_model", env=env)

# Create variables
# x has no upper bound, y is also unbounded
x = model.addVar(name="x", lb=0)
y = model.addVar(name="y", lb=0)

# Add a simple constraint
model.addConstr(x + y >= 10, name="constraint1")

# Set objective: Maximize x + 2*y (unbounded!)
model.setObjective(x + 2*y, GRB.MAXIMIZE)

print("\nModel created:")
print("  Variables: x >= 0, y >= 0")
print("  Constraint: x + y >= 10")
print("  Objective: Maximize x + 2*y")
print("\nThis model is UNBOUNDED because we can increase y infinitely!")
print("\n" + "=" * 70)

# Optimize the model
print("\nStep 1: Optimizing the model...")
print("=" * 70)
model.optimize()

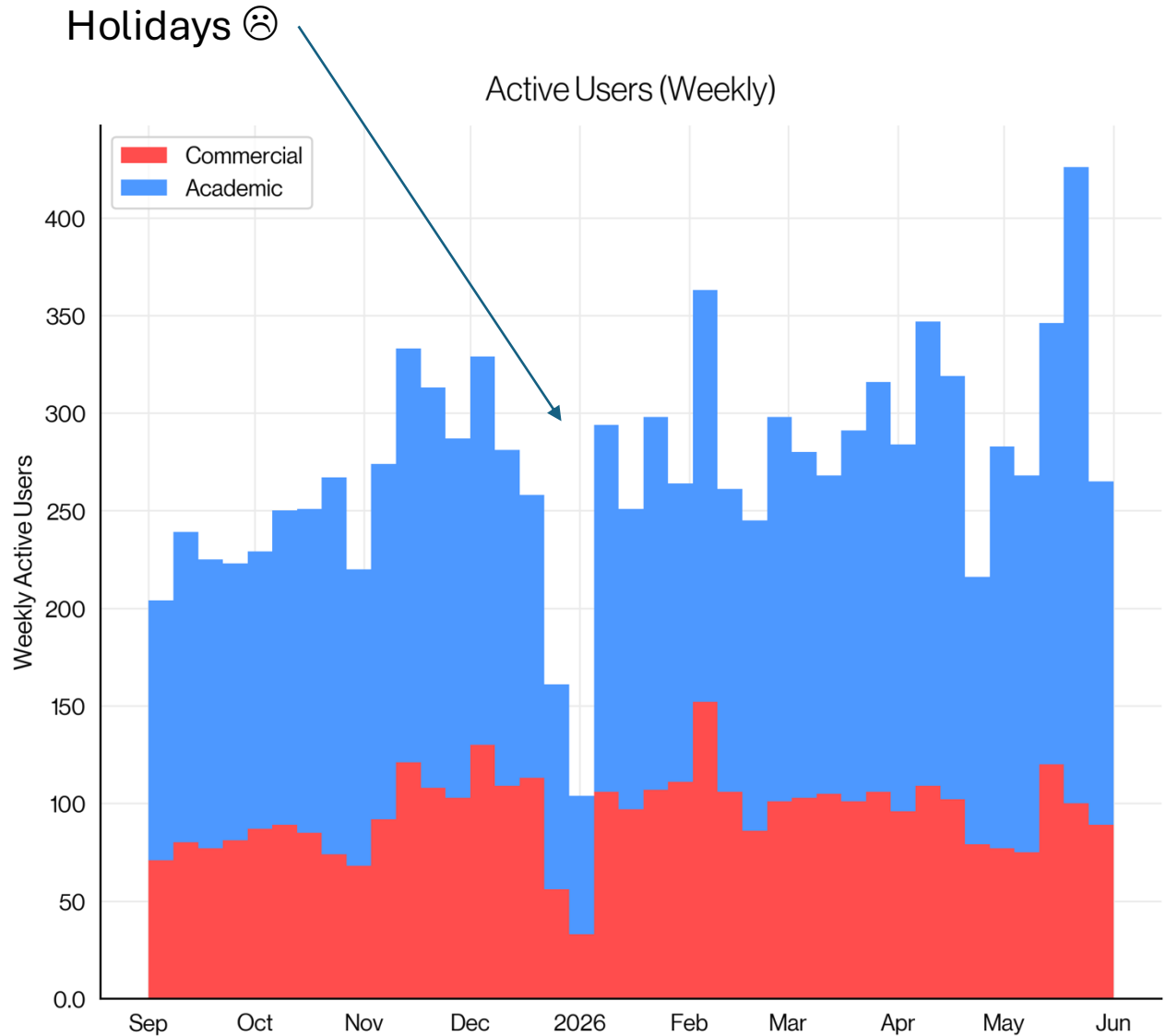
print("\n" + "=" * 70)
print(f"Optimization Status: {model.Status}")

if model.Status == GRB.UNBOUNDED:
    print("Status is UNBOUNDED")
elif model.Status == GRB.INF_OR_UNBD:
    print("Status is INF_OR_UNBD (Infeasible or Unbounded)")
```

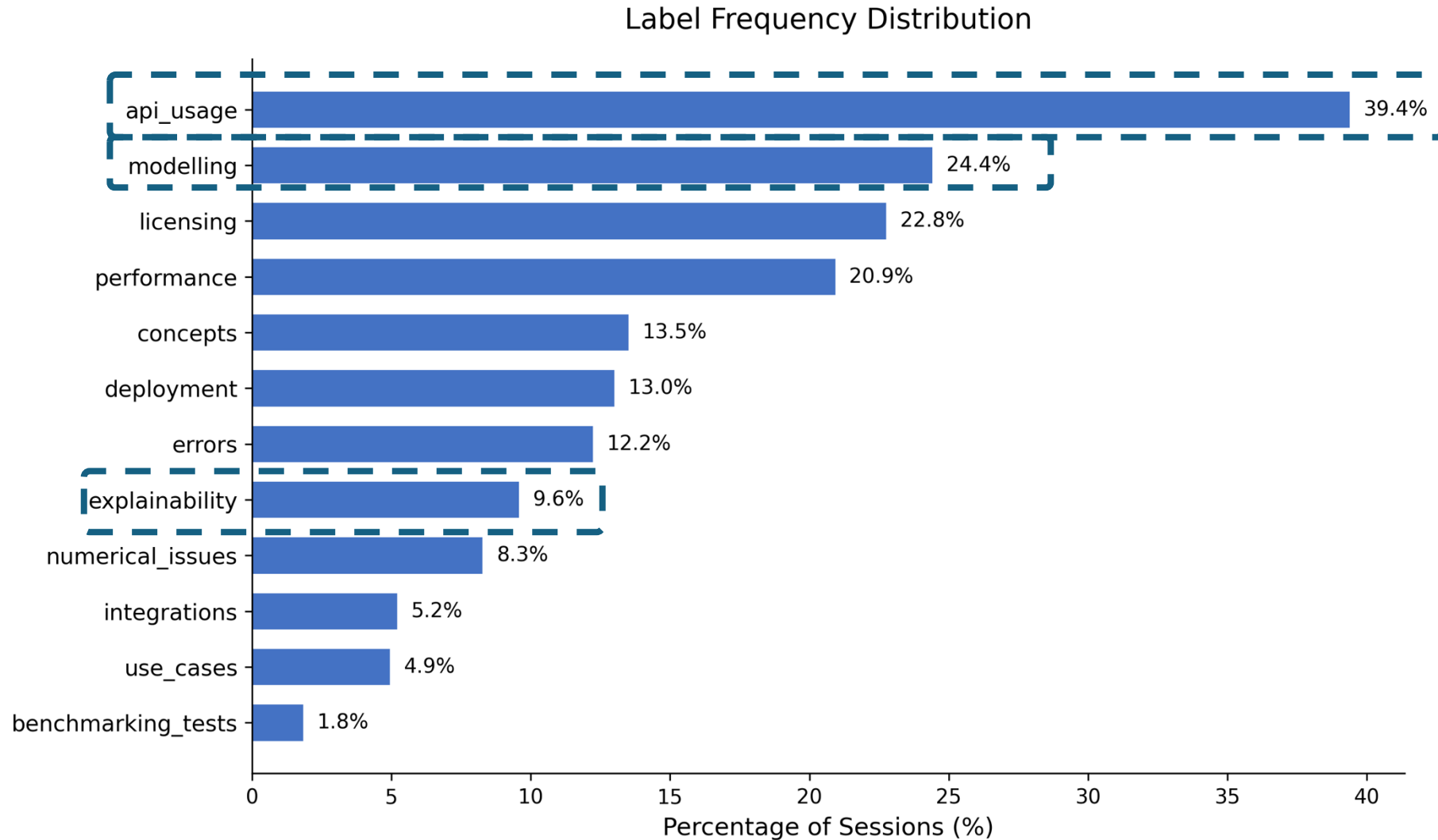
Gurobot metrics

July 2025 to today:

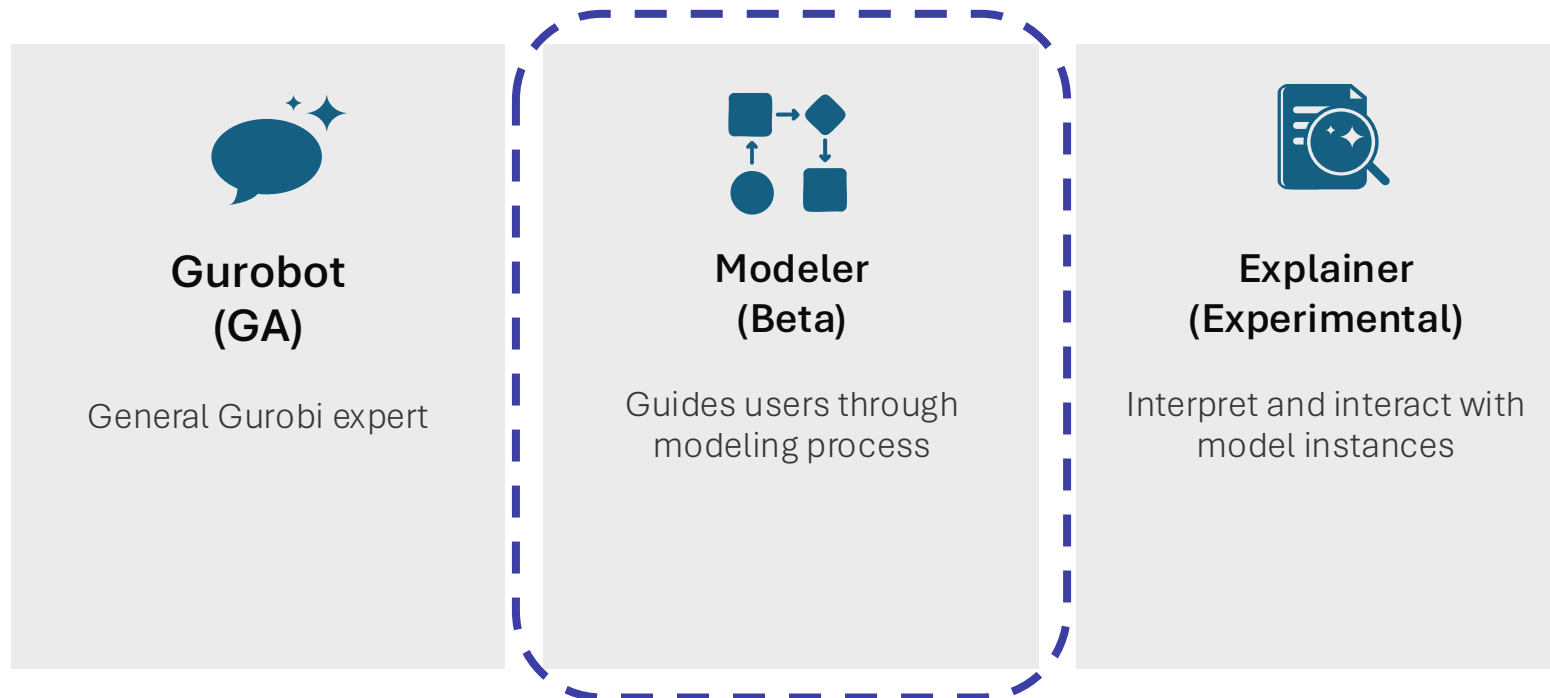
- 23k chats
- 7k unique users
- 32 languages
 - 75% of chats English



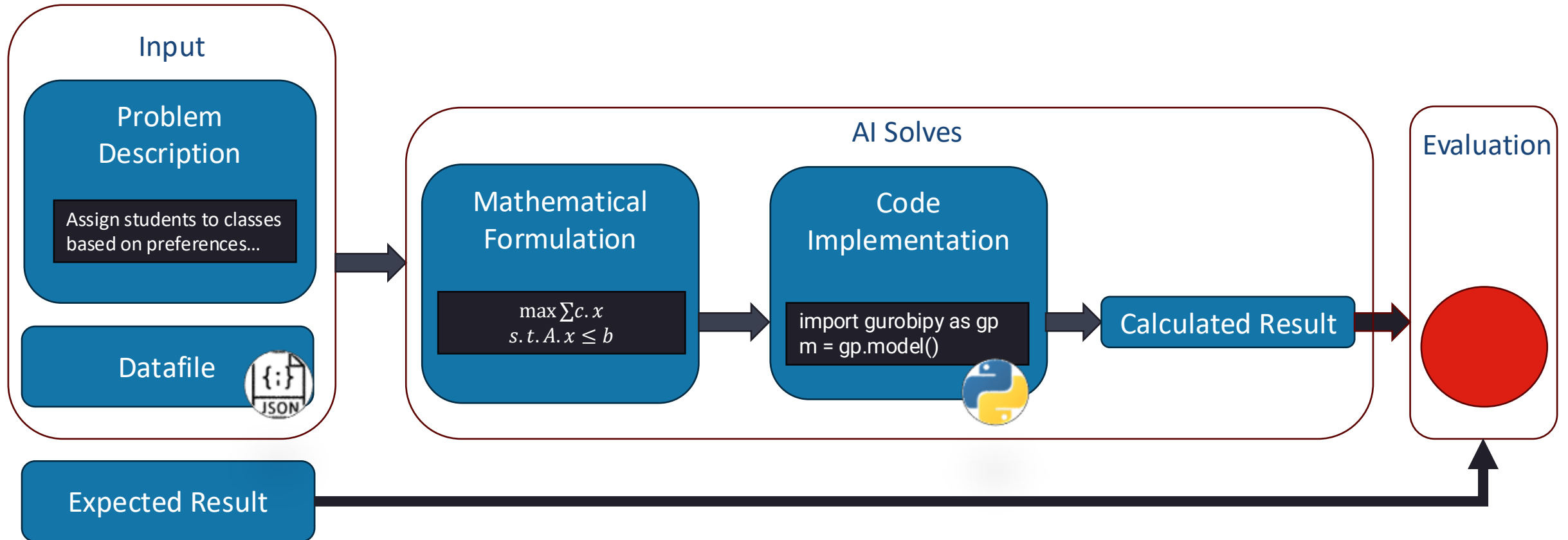
What people are using Gurobot for



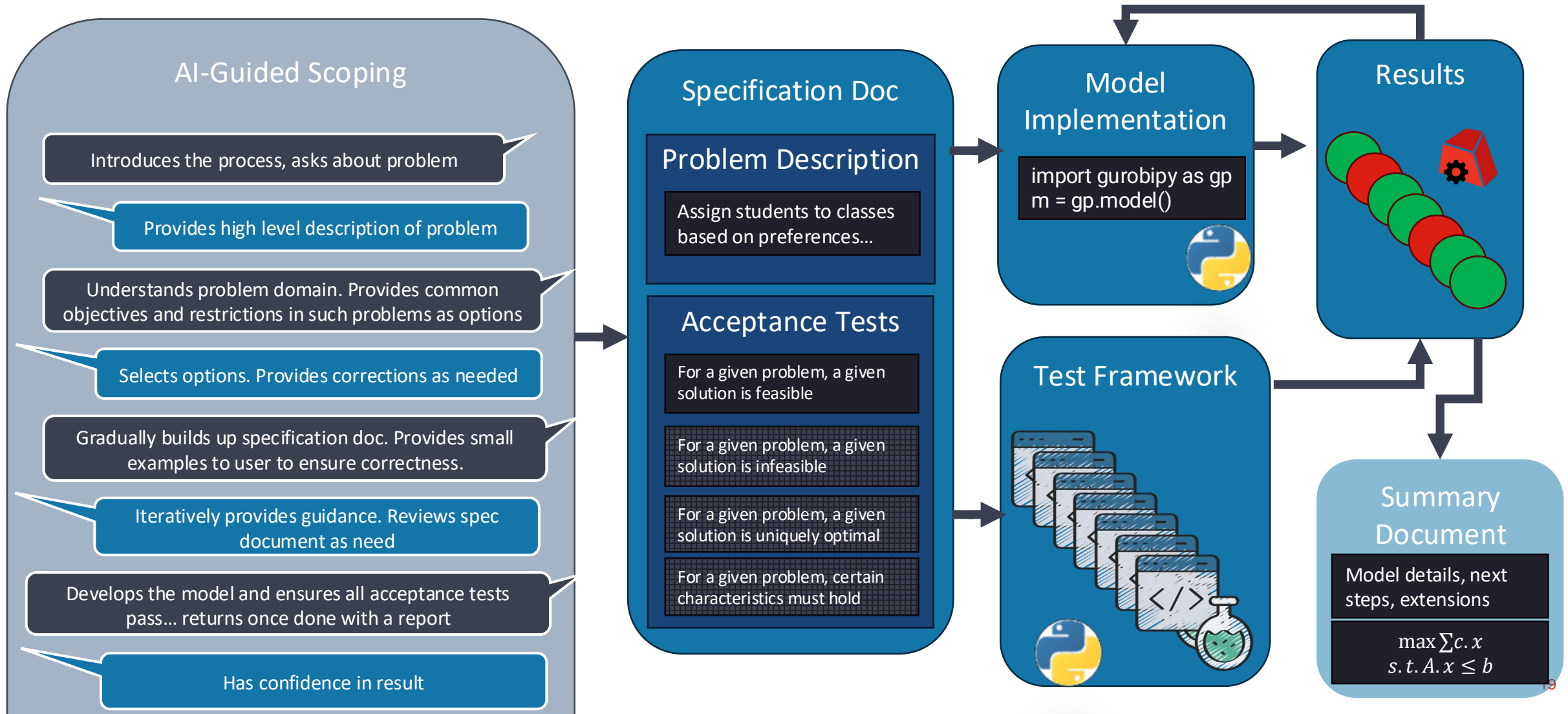
Overview of our agents



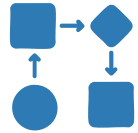
How GenAI for modeling is typically discussed



Spec-Based Test-Driven Modelling



Modeler



Guides user through a modeling workflow

A modeling assistant that guides users incrementally from a description of the problem to a structured specification, gurobipy implementation, unit tests and optional mathematical formulation



Conversational specification

Asks one clarifying question at a time to define objectives, constraints, inputs, outputs, and test scenarios



Business-first workflow

Keeps the user focused on the real-world problem



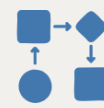
Grounded by acceptance tests

Generates pytest scenarios, runs them, fixes failures, flags inconsistencies



Iterative by design

As the problem changes, update the spec, and allow the AI to make the updates to the model and tests



Hi, I'm Modeler.

What business problem do you want to optimize?

I'll guide you through the rest, one question at a time, and deliver validated gurobipy code ready to run.

Type your message...





Help me write a time tabling model

Thank you for reaching out! Timetabling problems can take many forms, and I want to make sure I help you build the right model.

What type of timetabling problem are you trying to solve?

Please select from the following common scenarios (or describe your specific situation):

1. **Course/Class Timetabling** - Scheduling classes, teachers, and classrooms in educational institutions (universities, schools)
2. **Employee Shift Scheduling** - Creating work rosters/schedules for employees across different shifts with availability and requirement constraints
3. **Exam Timetabling** - Scheduling exams while avoiding conflicts and respecting room capacities
4. **Meeting/Room Scheduling** - Allocating meeting rooms and time slots for various meetings or events
5. **Transportation/Vehicle Timetabling** - Scheduling buses, trains, or other vehicles on routes at specific times
6. **Sports League Scheduling** - Creating game schedules for teams in a league
7. **Other** - Please describe your specific timetabling scenario

Which of these best describes your problem, or would you like to explain your specific use case?

Learn More:

- [Workforce Scheduling](#) ★★☆☆
- [Technician Routing and Scheduling Problem: Example solution to help telecommunications firms automate and improve their technician assignment, scheduling, and routing decisions](#) ★☆☆



Chat files

meeting_scheduling_specification.md (4.2 KB) ▾

Meeting/Room Scheduling Problem

Overview

The meeting/room scheduling problem aims to optimally assign meetings to available rooms and time slots to maximize the number of meetings that can be scheduled. The problem considers various constraints such as room capacity, participant availability, meeting durations, and time windows to ensure a feasible and efficient schedule.

Objective

Maximize the number of meetings that get successfully scheduled within the available rooms and time slots.

Constraints

C1: Room Conflict Constraint

A room can only host one meeting at a time. No two meetings can be assigned to the same room in the same time slot.

C2: Room Capacity Constraint

Each meeting requires a certain number of attendees, and the assigned room must have sufficient capacity to accommodate all attendees.

C3: Participant Availability Constraint

Meeting participants cannot attend multiple meetings simultaneously. If a person is required for

Perfect! I've defined the core input data structure with:

- **Meetings:** ID, list of attendees, number of attendees, and duration
- **Rooms:** ID and capacity
- **Time slots:** ID and time information

Now let's define some test scenarios. What situations should we test?

Please select the test scenarios that are important for your use case:

1. **Basic feasible case** - Simple scenario with enough rooms and time slots to schedule all meetings

2. **Oversubscribed resources** - More meetings than can be accommodated (rooms/time conflicts)

3. **Capacity constraints** - Meetings with varying sizes and rooms with different capacities

4. **Participant conflicts** - Same people needed in multiple meetings at overlapping times

5. **Long duration meetings** - Meetings that require multiple consecutive time slots

6. **Edge case with no meetings** - Empty meeting list

7. **Edge case with no rooms** - All meetings should be unscheduled

8. **All of the above** - Comprehensive test coverage

9. **Let me describe specific scenarios** - You have particular test cases in mind

Which test scenarios should we include?

Attachments:

 meeting_scheduling_specification.md





Just 1

Excellent! I've added the basic feasible test scenario to the specification.

Chat files

meeting_scheduling_specification.md (4.2 KB) 

Test Scenarios

T1: Basic Feasible Case

Description: Test with a simple scenario where there are enough rooms and time slots to schedule all meetings without conflicts. This validates that the model correctly schedules meetings when resources are sufficient.

Input Data:

• Meetings:

meeting_id	attendees	num_attendees	duration
M1	[P1, P2]	2	1
M2	[P3, P4]	2	1
M3	[P1, P3]	2	1

• Rooms:

room_id	capacity
R1	5
R2	5

• Time Slots:

slot_id	time_info
---------	-----------

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>

Perfect! I've successfully created your meeting room scheduling optimization model. Here's what has been completed:

Summary

Files Created:

1. **meeting_scheduling_specification.md** - Complete business specification
2. **meeting_scheduling.py** - Full implementation using Gurobi
3. **test_meeting_scheduling.py** - Comprehensive test suite

Model Features

Your model maximizes the number of scheduled meetings while respecting:

1. **Room Conflicts** - One meeting per room per time slot
2. **Room Capacity** - Meetings only assigned to rooms large enough
3. **Participant Availability** - People can't be in two meetings at once
4. **Meeting Duration** - Multi-slot meetings get consecutive time slots
5. **Time Windows** - Meetings scheduled within available slots

Test Results

All 5 test cases passed successfully:

- ✓ T1: Basic feasible case (from specification)
- ✓ Large meeting capacity constraint test
- ✓ Multi-slot duration test
- ✓ Participant conflict test
- ✓ Empty meetings list edge case

Example Output

The example demonstrates scheduling 4 meetings across 2 rooms and 3 time slots, successfully scheduling all meetings without conflicts.

Documentation ▾

Downloads ▾

Support ▾



 Steven Edwards ▾

Chat files

test_meeting_scheduling.py (8.4 KB) ▾

```
"""
Unit tests for Meeting Room Scheduling Optimizer
"""

import pytest
import gurobipy as gp
from meeting_scheduling import MeetingSchedulingOptimizer

@pytest.fixture
def gurobi_env():
    """Create a Gurobi environment for testing."""
    with gp.Env(empty=True) as env:
        env.setParam('OutputFlag', 0) # Suppress output during tests
        env.start()
        yield env

class TestMeetingScheduling:
    """Test cases for meeting scheduling optimization."""

    def test_T1_basic_feasible_case(self, gurobi_env):
        """
        T1: Basic Feasible Case

        Test with a simple scenario where there are enough rooms and time slots
        to schedule all meetings without conflicts.
        """
        # Input data: 4 meetings, 2 rooms, 3 time slots
        meetings = [
            {'id': 'M1', 'attendees': ['Alice', 'Bob'], 'num_attendees': 2, 'duration': 1},
            {'id': 'M2', 'attendees': ['Charlie', 'David'], 'num_attendees': 2, 'duration': 1},
            {'id': 'M3', 'attendees': ['Alice', 'Charlie'], 'num_attendees': 2, 'duration': 1},
            {'id': 'M4', 'attendees': ['Bob', 'Eve'], 'num_attendees': 2, 'duration': 1},
        ]

        rooms = [
            {'id': 'R1', 'capacity': 4},
            {'id': 'R2', 'capacity': 2},
        ]
```

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Show me the maths

The mathematical formulation has been saved to `meeting_scheduling_formulation.md`. Here is a summary of the full model:

Sets and Indices

- M - set of meetings
- R - set of rooms
- T - set of ordered time slots
- P - set of participants, where $A_m \subseteq P$ are the attendees of meeting m

Parameters

- d_m - duration of meeting m (in time slots)
- n_m - number of attendees for meeting m
- c_r - capacity of room r

Decision Variable

A single binary variable drives the entire model:

$x_{m,r,t} = 1$ if meeting m starts in room r at time slot t , 0 otherwise.

Objective

$$\max \sum_{m \in M} \sum_{r \in R} \sum_{t \in T} x_{m,r,t}$$

Constraints

Chat files

meeting_scheduling_formulation.md (5.4 KB)

4. Objective Function

Maximize the total number of scheduled meetings:

$$\max \sum_{m \in M} \sum_{r \in R} \sum_{t \in T} x_{m,r,t}$$

5. Constraints

C1: Meeting Assignment - Each meeting is scheduled at most once

Each meeting can be assigned to at most one room-time slot combination:

$$\sum_{r \in R} \sum_{t \in T} x_{m,r,t} \leq 1 \quad \forall m \in M$$

C2: Room Conflict - A room can host at most one meeting at any time slot

For each room r and each time slot t (at position index t_{idx}), all meetings that would occupy that slot must not exceed one. A meeting m started at time slot with index s_{idx} occupies time slot t_{idx} if:

$$s_{idx} \in [\max(0, t_{idx} - d_m + 1), t_{idx}]$$

The constraint is:

$$\sum_{m \in M} \sum_{s_{idx} = \max(0, t_{idx} - d_m + 1)}^{t_{idx}} x_{m,r,T[s_{idx}]} \leq 1 \quad \forall r \in R, \forall t_{idx} \in \{0, \dots, |T| - 1\}$$

C3: Room Capacity - Room must accommodate all attendees

Modeler helps you build a correct model.

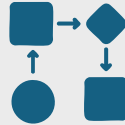
Does not guarantee it is the most performant model.

Overview of our agents



Gurobot (GA)

General Gurobi expert



Modeler (Beta)

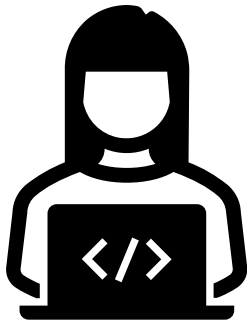
Guides users through
modeling process



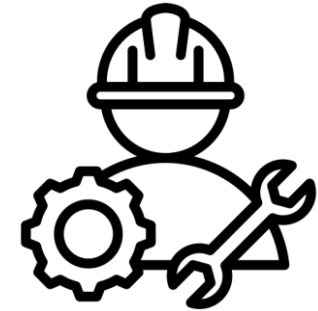
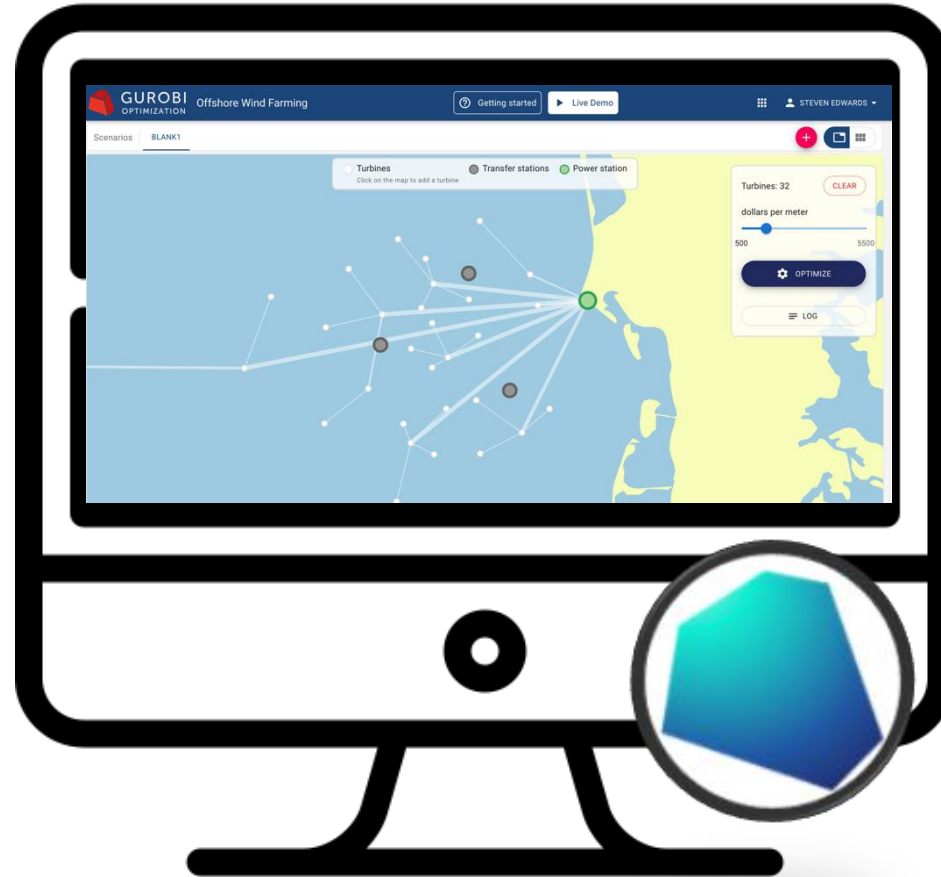
Explainer (Experimental)

Interpret and interact with
model instances

End Users are often not optimization experts



Built by Developer



Used by End User

Doesn't know Gurobi even exists.

End Users can have MANY questions

Infeasibility – Explain

```
model.computeIIS()
```

- Why is my model infeasible?
- Why can't my supply plan meet demand?
- Why can't I generate a feasible production schedule?
- Why is this solution considered infeasible?

```
# Duality Information
for v in model.getVars():
    v.RC
for c in model.getConstrs():
    c.Pi
```

Sensitivity Analysis

- Which constraints or bounds should I relax to most improve my objective?
- Where should I invest more money to expect larger profits?
- How far can resource availability drop before we switch to a different plan?

Infeasibility – Repair

```
model.feasRelax(...)
```

- How can I minimally adjust my constraints to make my model feasible?
- If I allow partial deliveries, would the model become feasible?
- Which due dates are too tight and must be extended?

```
# Simple tricks
modified = model.copy()
modified.addConstr(...)
modified.optimized()
```

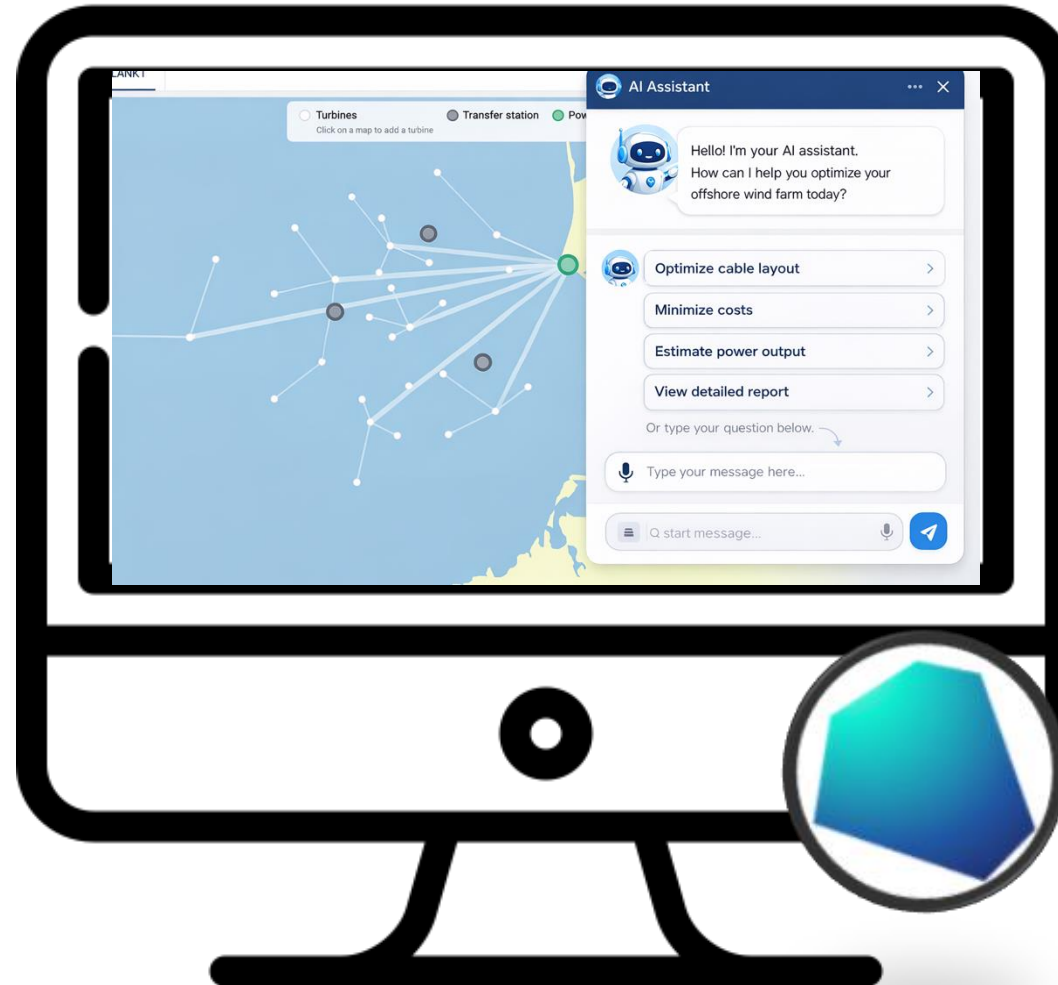
Counterfactuals

- Why is my truck not fully loaded?
- Why did the solver prefer this solution over the alternative I expected?
- If I force product P onto machine M, which jobs are displaced?



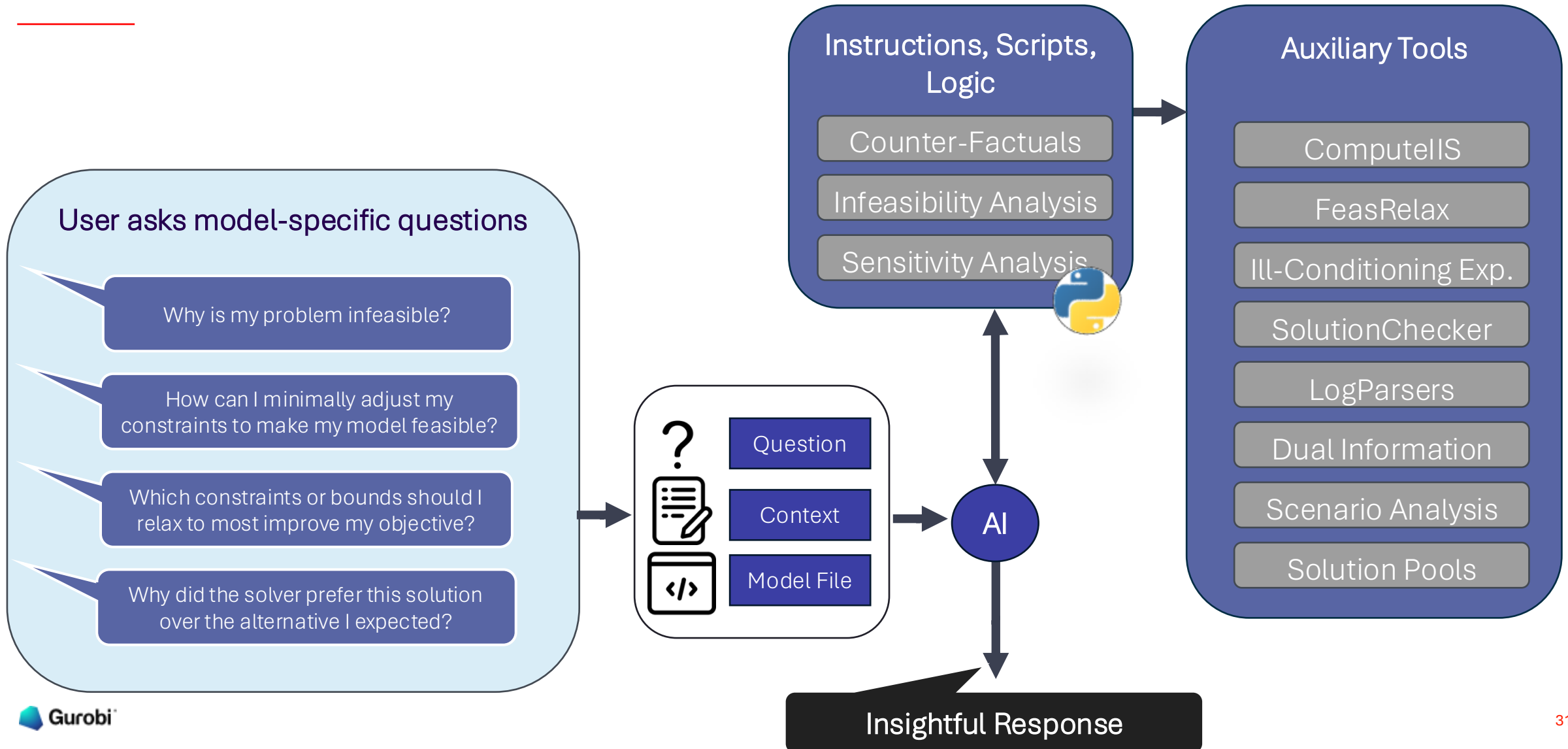
New paradigm of decision support tools emerging

GenAI used as
translation layer
between end-user and
model



Gurobi still used to
solve the optimization
problems

Ground AI responses using tools



Explainer



Making advanced model analysis more accessible

Explainer helps users interpret model instances by combining Gurobi's analytical capabilities with business context



Unlock Gurobi diagnostics

Guided access to Gurobi analysis tools



Explain results in domain context

Clarify what outputs mean in the language of the domain



Focused workflows

Start with infeasibility analysis, feasibility restoration, and sensitivity analysis



Connect to next steps

Suggest practical actions based on analysis



Hi, I'm Explainer.

What should I help you analyze first?

-  **Infeasibility Diagnosis**
-  Feasibility Restoration
-  Sensitivity Analysis

What infeasible problem do you need help understanding?

Upload your infeasible model (either the model file or the **IIS in ILP format** [↗]) – and any supporting files, such as code, documentation, data files, or modeling notes. I'll identify the root causes of your infeasibility with concrete fixes.

[Learn more](#) [↗]

1 Upload your model and supporting files

REQUIRED

Choose a file or drag it here.

 **BROWSE FILES**

2 Ask a question and add additional context

OPTIONAL

Helpful context includes your ultimate goal, relevant assumptions, and notes about your model, such as naming conventions.

EXAMPLES YOU COULD ASK:

My Gurobi model is infeasible, what is causing the conflict?

↑ Use

 **SEND**



Hi, I'm Explainer.

What should I help you analyze first?

-  Feasibility Diagnosis
-  **Feasibility Restoration**
-  Sensitivity Analysis

What infeasible problem do you need help repairing?

Upload your infeasible model — and any supporting files, such as code, documentation, data files, or modeling notes. I'll help you identify which constraints and variables bounds to relax, and by how much, to restore feasibility.

[Learn more ↗](#)

1 Upload your model and supporting files

REQUIRED

Choose a file or drag it here.

 [BROWSE FILES](#)

2 Ask a question and add additional context

OPTIONAL

Helpful context includes your ultimate goal, relevant assumptions, and notes about your model, such as naming conventions.

EXAMPLES YOU COULD ASK:

- What is the minimum total understaffing needed for feasibility?  Use
- How can we spread unmet demand across regions?  Use
- What is the fewest requirements we need to relax?  Use

 SEND

Start with the Workflow for Explaining Infeasibility. why is my model infeasible?

Files Uploaded:

 model.lp



 README.1.md



The full report has been saved to `infeasibility_report_moodel.md` in your workspace. Here is a summary of the findings, Steven.

Why Your Model Is Infeasible

The IIS contains **19 constraints and 9 bounds**. The root cause is a conflict between three elements that pull in opposite directions:

The triggering constraint: `Minimum_Premium >= 20,000` The README hints at this directly – *"someone quietly slipped in a new requirement during a marketing push."* Before this constraint was added, the model was likely feasible.

The structural problem:

- Every cream fraction produced (light, medium, heavy) has a quality index **below** the premium drink's required quality of 94. This means every barrel of cream dilutes the blend below target.
- The only ingredients that boost quality above 94 are **protein concentrate** (quality 115) and **fermented concentrate** (quality 105). Their production is capped by two bottleneck units:
 - Protein concentration unit: 10,000 barrels/day input → max ~5,720 barrels of protein concentrate
 - Secondary processing unit: 8,000 barrels/day input → max ~1,984 barrels of fermented concentrate
- At maximum milk supply, the plant produces **9,400 barrels of heavy cream** (quality 70, a deficit of 24 points vs. the 94 threshold). Routing it to premium requires a quality boost of $24 \times 9,400 = 225,600$ surplus units, but the maximum available is only 141,944. The shortfall is approximately **2.4×** what the plant can supply.

Chat files

infeasibility_report_moodel.md (10.1 KB)

Infeasibility Analysis Report

File Analyzed: moodel.lp
Analysis Date: 2026-06-10
Generated For: Steven Edwards

1. IIS Overview

The Irreducible Inconsistent Subsystem (IIS) computed by Gurobi contains **19 constraints and 9 variable bounds**. All constraints and variables carry descriptive names, making the IIS straightforward to interpret.

The IIS spans the following constraint groups:

Group	Constraints Involved
Capacity limits	Protein_conc_cap, Secondary_proc_cap
Separation yield equations	Splitting_separation[Light_cream], [Medium_cream], [Heavy_cream], [Skim_milk]
Protein and fermented concentrate production	Splitting_protein_concentration, Splitting_secondary_processing[Fermented_concentrate]
Cream and intermediate	Continuity_creams[Light_cream], [Medium_cream], [Heavy_cream], Fixed_proportion_skim_for_base_mix,



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yes please relax the quality of the premium drink

The repaired model is confirmed feasible with an optimal objective value of **196,213.09**.

Here is what fea^sre^lax computed:

Feasibility Restoration Result

What was relaxed

The Quality_{tol_premium_drink} constraint is a blending quality constraint of the form:

$$\begin{aligned} &(\text{sum of quality contributions from all ingredients routed to premium drink}) \geq 94 \\ &x \text{ (total premium drink volume)} \end{aligned}$$

It is written in standard form with everything moved to one side, giving a RHS of **0.0**.

fea^sre^lax decreased that RHS to **-12,371.28**, which is the minimum relaxation required to restore feasibility.

What this means in practice

The effective relaxation corresponds to **lowering the premium drink quality threshold**. The minimum reduction needed is approximately **12,371 quality-weighted barrel-units**. Given the 20,000-barrel minimum premium volume, this translates to a quality index reduction of roughly:

$$12,371 / 20,000 = \mathbf{0.62 \text{ quality index points}}$$

(i.e., the threshold drops from 94 to approximately **93.38**)

This is the minimum required. Meeting or slightly exceeding this relaxation keeps the model feasible.

In plain terms

The premium drink quality specification of 94 needs to be reduced to approximately **93.4 or lower** in order for the plant's current processing capacities to satisfy the 20,000-barrel

Chat files

moodel_repaired.lp (5.1 KB) ▾

\ LP format - for model browsing. Use MPS format to capture full model detail.

Maximize

$$\begin{aligned} &7 \text{ end_prod[Premium_drink]} + 6 \text{ end_prod[Standard_drink]} \\ &+ 4 \text{ end_prod[Cooking_cream]} + 3.5 \text{ end_prod[Dairy_base_mix]} \\ &+ 1.5 \text{ end_prod[Butter_solids]} \end{aligned}$$

Subject To

$$\text{Separation_cap: milk_in[1] + milk_in[2] \leq 45000}$$

$$\begin{aligned} &\text{Protein_conc_cap: creams_to_protein_concentrate[Light_cream]} \\ &+ \text{creams_to_protein_concentrate[Medium_cream]} \\ &+ \text{creams_to_protein_concentrate[Heavy_cream]} \leq 10000 \end{aligned}$$

$$\begin{aligned} &\text{Secondary_proc_cap: milk_bases_to_secondary_processing[Skim_milk]} \\ &+ \text{milk_bases_to_secondary_processing[Concentrated_skim]} \leq 8000 \end{aligned}$$

$$\begin{aligned} &\text{Splitting_separation[Light_cream]: } 0.1 \text{ milk_in[1] + } 0.15 \text{ milk_in[2]} \\ &- \text{sep_prod[Light_cream]} = 0 \end{aligned}$$

$$\begin{aligned} &\text{Splitting_separation[Medium_cream]: } 0.2 \text{ milk_in[1] + } 0.25 \text{ milk_in[2]} \\ &- \text{sep_prod[Medium_cream]} = 0 \end{aligned}$$

$$\begin{aligned} &\text{Splitting_separation[Heavy_cream]: } 0.2 \text{ milk_in[1] + } 0.18 \text{ milk_in[2]} \\ &- \text{sep_prod[Heavy_cream]} = 0 \end{aligned}$$

$$\begin{aligned} &\text{Splitting_separation[Skim_milk]: } 0.12 \text{ milk_in[1] + } 0.08 \text{ milk_in[2]} \\ &- \text{sep_prod[Skim_milk]} = 0 \end{aligned}$$

$$\begin{aligned} &\text{Splitting_separation[Concentrated_skim]: } 0.2 \text{ milk_in[1] + } 0.19 \text{ milk_in[2]} \\ &- \text{sep_prod[Concentrated_skim]} = 0 \end{aligned}$$

$$\begin{aligned} &\text{Splitting_separation[Residual_solids]: } 0.13 \text{ milk_in[1] + } 0.12 \text{ milk_in[2]} \\ &- \text{sep_prod[Residual_solids]} = 0 \end{aligned}$$

Splitting_protein_concentration:

$$\begin{aligned} &0.6 \text{ creams_to_protein_concentrate[Light_cream]} \\ &+ 0.52 \text{ creams_to_protein_concentrate[Medium_cream]} \\ &+ 0.45 \text{ creams_to_protein_concentrate[Heavy_cream]} - \text{protein_concentrate} \\ &= 0 \end{aligned}$$

Splitting_secondary_processing[Fermented_concentrate]:

$$\begin{aligned} &0.28 \text{ milk_bases_to_secondary_processing[Skim_milk]} \\ &+ 0.2 \text{ milk_bases_to_secondary_processing[Concentrated_skim]} \\ &- \text{secondary_prods[Fermented_concentrate]} = 0 \end{aligned}$$

Splitting_secondary_processing[Processed_base]:

$$\begin{aligned} &0.68 \text{ milk_bases_to_secondary_processing[Skim_milk]} \\ &+ 0.75 \text{ milk_bases_to_secondary_processing[Concentrated_skim]} \\ &- \text{secondary_prods[Processed_base]} = 0 \end{aligned}$$

Splitting_butter_solids: - end_prod[Butter_solids]

$$+ 0.5 \text{ residual_used_for_butter} = 0$$

Blending_premium_drink: - end_prod[Premium_drink]

$$+ \text{to premium drink[Light cream]} + \text{to premium drink[Medium cream]}$$

Which tank should I increase capacity of to improve my profit

The sensitivity analysis is complete. Here is the answer, Steven.

Which Tank Capacity to Increase

Of all the binding constraints in the model, two correspond directly to processing unit (tank) capacities:

Tank	Shadow Price	Allowable RHS Increase
Protein concentration unit (Protein_conc_cap)	74.50 USD per barrel/day	up to +179 barrels/day
Secondary processing unit (Secondary_proc_cap)	16.35 USD per barrel/day	up to +857 barrels/day

Recommendation: Expand the Protein Concentration Unit

The protein concentration unit has a shadow price of 74.50 USD per barrel per day. For each additional barrel per day of input capacity, profit increases by 74.50 USD. That is approximately 4.6 times more impactful than expanding the secondary processing unit, which has a shadow price of 16.35 USD per barrel per day.

Important Caveat on Range

The shadow price of 74.50 USD holds only within the current optimal basis. That basis remains valid for capacity increases of up to approximately 179 barrels per day, taking the unit from 10,000 to at most 10,179 barrels per day. Beyond that threshold the basis changes and a fresh sensitivity run would be needed. The secondary processing unit has a wider valid range, remaining stable for increases of up to 857 barrels per day.

Summary

Chat files

gurobi.log (6.4 KB)

Gurobi 13.0.2 (armlinux64, Python) logging started Wed Jun 10 13:07:23 2026

Set parameter LogFile to value "gurobi.log"
Restricted license – for non-production use only – expires 2027-11-29
Read LP format model from file /tmp/moodel.lp
Reading time = 0.00 seconds
: 30 rows, 36 columns, 107 nonzeros
Set parameter TimeLimit to value 60
Gurobi Optimizer version 13.0.2 build v13.0.2rc1 (armlinux64 – "Debian GNU/Linux 13 (trixie)")

CPU model: ARM64
Thread count: 2 physical cores, 2 logical processors, using up to 2 threads

Non-default parameters:
TimeLimit 60

Iteration	Objective	Primal Inf.	Dual Inf.	Time
0	8.3125000e+30	2.151250e+30	8.312500e+00	0s

IIS computed: 19 constraints and 9 bounds
IIS runtime: 0.00 seconds (0.00 work units)

Gurobi 13.0.2 (armlinux64, Python) logging started Wed Jun 10 13:13:23 2026

Set parameter LogFile to value "gurobi.log"
Restricted license – for non-production use only – expires 2027-11-29
Read LP format model from file /tmp/moodel.lp
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Summary

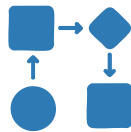
Summary



Intelligence Hub
Home of our AI tools



Gurobot
AI-powered optimization
assistant, escalate to a human



Modeler (Beta)
Guides users through modeling
process



Explainer (Experimental)
Allows users to ask questions
about model instances

Gurobi Intelligence Hub

Welcome to the Gurobi Intelligence Hub—your AI command center for mathematical optimization.

The Gurobi Intelligence Hub is a web application that provides AI-powered agents to help you work more effectively with Gurobi products. The Intelligence Hub currently offers three specialized agents:

- **Gurobot:** An AI assistant that delivers instant, expert-level Gurobi support with personalized, context-aware answers, code snippets, and documentation links backed by Gurobi’s knowledge base.
- **Explainer:** An optimization analysis agent that diagnoses infeasibility, proposes feasibility repairs, and runs sensitivity analysis on your models, grounded in Gurobi’s curated knowledge base and optimization expertise.
- **Modeler:** A modeling assistant that helps you co-develop a clear, plain-language optimization specification with AI, then turns that specification into a complete optimization project—including a mathematical formulation, `gurobipy` implementation, and validated unit tests.

The Intelligence Hub also provides an MCP server that runs locally on your machine, allowing you to access Gurobot, Explainer, and Modeler directly from MCP-compatible tools such as IDE extensions and AI coding assistants. API keys are used to generate a license file that authenticates the MCP server with the Intelligence Hub backend.

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- **Free for all users:** currently 2 million tokens per day.
- **Fresh knowledge, reindexed daily:** new knowledge base articles are available immediately.

01.

Gurobot

Constantly improving the articles that we use to address AI blind spots. Easy escalation.

02.

Explainer

Many customers trying to do this individually and this is not trivial at all. Would love to be able to contribute a tool that our customers can use directly

03.

Modeller

Simple workflow that we find significantly improves the models being generated.

Try it out! intelligence.gurobi.com



Thank You

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